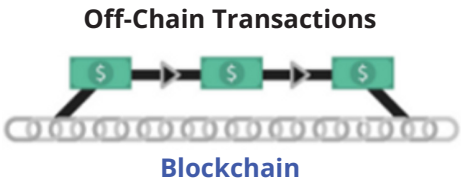


The Lightning Network is a scaling **approach** to ***Bitcoin***. It's really about moving a lot of ***Bitcoin*** transactions **out of the blockchain and into private channels** between users, but still counting on the blockchain's security.

If using the lightning network, only **three types of transactions** need to be broadcast to the *blockchain*.

The transaction that **opens** a channel.

The transaction that **closes** a channel.

**Disputes** that need to be settled.

| Payment Network | Bitcoin Network | Lightning Network |
|---|---|---|
| **Definition** | A decentralized digital network that uses cryptography to secure financial transactions. | A second layer payment protocol that operates on top of the ***Bitcoin blockchain***, enabling faster and cheaper transactions. |
| **Advantages** | - Decentralized and secure - No chargebacks or fraud - Can be used anonymously - Global Acceptance | - Faster and cheaper transactions - Increased scalability - Off-chain transactions do not clog the blockchain |
| **Disadvantages** | - Slow transaction times - High fees for certain types of transactions- Complex for beginners | - Requires trust in the channel operators - Still experimental and not widely adopted - Requires on-chain transaction to open and close channels |

### *6.3 The Lightning Network*

***Bitcoin*** is known for its unalterable public ledger, but it may not be the best choice for everyday transactions like buying coffee. The process of broadcasting these transactions to many nodes and storing them in a shared database can be slow and cumbersome. For personal or private transactions, it's better to use peer-to-peer payment channels.

A better solution is a layered approach to scaling, such as the combination of ***Bitcoin*** and Lightning Networks. This allows users to pick the layer that fits their needs. ***Bitcoin*** is a digital currency that is decentralized, while the Lightning Network provides quick, cheap, and confidential payments.